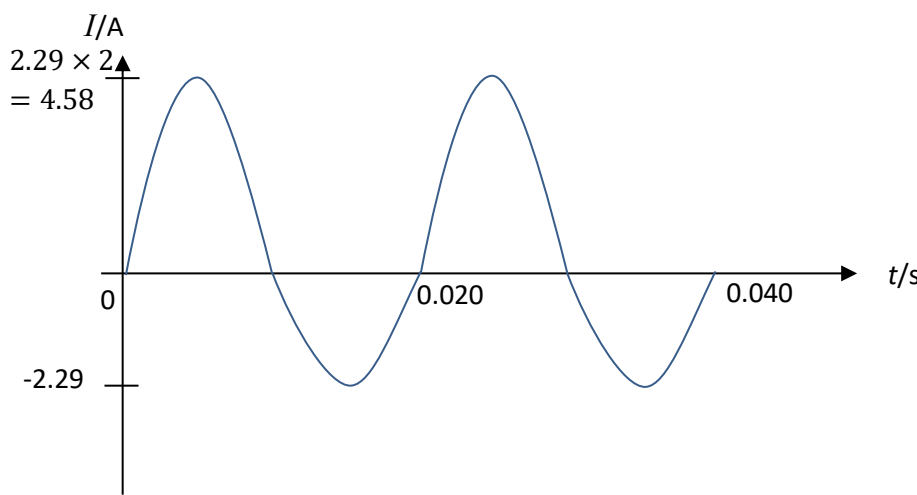


5	(a)	(i)	$V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{170}{\sqrt{2}} = 120 \text{ V}$	1
		(ii)	$\omega = 2\pi/T = 314$ $T = 0.0200 \text{ s}$	1
	(b)	(i)	$\frac{V_S}{V_P} = \frac{N_S}{N_P}$ $\frac{V_S}{170} = \frac{3500}{2000}$ $V_S = 298 \text{ V}$	1 1
		(ii)	$V_S = I_S R$ $298 = I_S(130)$ $I_S = 2.288 \text{ A}$ $\frac{I_P}{I_S} = \frac{N_S}{N_P} = \frac{3500}{2000}$ For correct working above: $I_P = 4.00 \text{ A}$	1 1
	(b)	 Shape of graph: 1 mark. Correct labels: 1 mark		